PRACTICE: Chemical Nomenclature **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) Name the following ionic compounds:

a. LiCl

1a

b. $Mg(OH)_2$

1b

c. K_3P

1c

d. Fe_2O_3

1d

e. FeO

1e

f. $ZnCl_2$

1f

g. $AgNO_3$

1g

h. NH_4Cl

1h

i. $CuCl_2$

1i

j. $SnCl_2$

1j

k. PbO_2

1k

l. $AlCl_3$

1l

m. $PbSO_4$

1m

n. $Mg_3(PO_4)_2$

1n

o. Na_2CO_3

1o

p. $NaHCO_3$

1p

q. KCN

1q

r. $KMnO_4$

1r

2) Write ionic formulas for the following compounds:

a. Sodium acetate

2a

b. Tin(II) chloride

2b

c. Calcium hydroxide

2c

d. Zinc sulfite

2d

e. Ammonium sulfate

2e

f. Manganese(II) hypochlorite

2f

g. Copper(I) nitrite

2g

h. Silver cyanide

2h

i. Nickel(II) chlorite

2i

j. Lead(II) nitrate

2j

k. Sodium chloride

2k

l. Lithium fluoride

2l

m. Potassium sulfide

2m

n. Aluminum oxide

2n

3) Write ionic formulas for the following compounds:

a. Calcium bromide

3a

b. Potassium phosphide

3b

c. Iron(III) oxide

3c

d. Iron(II) oxide

3d

e. Magnesium hydroxide

3e

f. Nickel(III) sulfate

3f

4) Write chemical formulas for the following hydrates:

a. Magnesium perchlorate hexahydrate

4a

b. Calcium chromate dihydrate

4b

c. Aluminum chloride hexahydrate

4c

d. Copper(II) sulfate pentahydrate

4d

e. Iron(III) chloride hexahydrate

4e

f. Barium chloride dihydrate

4f

PRACTICE: Chemical Nomenclature **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

5) Name the following hydrates:

a. $CoCl_2 \cdot 6H_2O$

5a

b. $SnCl_2 \cdot 2H_2O$

5b

c. $Ba(OH)_2 \cdot 8H_2O$

5c

d. $CaSO_4 \cdot 2H_2O$

5d

e. $MgSO_4 \cdot 7H_2O$

5e

6) Name the following covalent compounds:

a. CO

6a

b. CO_2

6b

c. H_2O

6c

d. CCl_4

6d

e. N_2O_3

6e

f. SiO_2

6f

g. N_2O

6g

h. CBr_4

6h

i. SO_2

6i

j. S_2Cl_2

6j

7) Write molecular formulas for the following compounds:

a. Xenon hexafluoride

7a

b. Tetranitrogen tetroxide

7b

c. Boron trifluoride

7c

d. Carbon tetrabromide

7d

e. Dicarbon tetrafluoride

7e

f. Nitrogen tribromide

7f

g. Dinitrogen tetrasulfide

7g

h. Oxygen difluoride

7h

i. Dinitrogen pentoxide

7i

j. Tetraphosphorus decoxide

7j

8) Write chemical formulas for the following compounds:

a. Carbon monoxide

8a

b. Potassium sulfide

8b

c. Potassium permanganate

8c

d. Nitrogen trichloride

8d

e. Nickel(II) sulfate

8e

f. Sodium acetate

8f

g. Magnesium iodate

8g

h. Sulfur trioxide

8h

i. Calcium carbonate

8i

j. Silver sulfide

8j